



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

SEMESTER 2 2023/2024

SECI1143 – KEBARANGKALIAN STATISTIK & ANALISIS DATA (PROBABILITY & STATISTICAL DATA ANALYSIS)

SECTION 03

ASSIGNMENT 2 – CHAPTER 3 &4

LECTURER: DR. NOR HAIZAN BT MOHAMED RADZI

STUDENT NAME	MATRIC NO
LAU YEE WEN	A23CS0099
GUI KAH SIN	A23CS0080
CHERYL CHEONG KAH VOON	A23CS0060
TAN SZE QING	A23CS5048

Class Interval	midpoint, x	frequency, f	fx	Cumulative frequency
$150 \leq x < 160$	155	12	1860	12
$160 \leq x < 170$	165	20	3300	32
$170 \leq x < 180$	175	5	875	37
$180 \leq x < 190$	185	3	555	40
		$\Sigma f = 40$	$\Sigma fx = 6590$	

(a) Mean, $\bar{x} = \frac{\Sigma fx}{\Sigma f}$

$$= \frac{6590}{40}$$

$$= 164.75$$

(b) $N = 40 \div 2$
 $= 20$

$L = 160$ Median = $L + \left[\frac{\frac{N}{2} - cf_p}{f_{med}} \right] (w)$
 $N = 40$

median class = $160 \leq x < 170$ $cf_p = 12$ $= 160 + \left[\frac{\frac{40}{2} - 12}{20} \right] (10)$
 $w = 10$
 $f_{med} = 20$ $= 164$

(c) modal class = $160 \leq x < 170$

$L = 160$ mode = $L + h \times \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right]$
 $f_1 = 20$
 $f_0 = 12$ $= 160 + 10 \times \left[\frac{20 - 12}{2(20) - 12 - 5} \right]$
 $f_2 = 5$
 $h = 10$ $= 163.478$

(d) Modal Class = $160 \leq x < 170$

Question 2

15, 80, 82, 85, 85, 85, 88, 90, 90, 92

a) $\bar{x} = \frac{\sum x}{n}$

$$= \frac{15 + 80 + 82 + 85 + 85 + 88 + 90 + 90 + 92 + 85}{10}$$
$$= 85.2$$

5th observation = 85

6th observation = 85

$$\text{Median} = \frac{1}{2}(85 + 85)$$
$$= 85$$

Mode = 85

- b) The mean score of 85.2 suggests that the overall performance of the students on the quiz was quite good, with the average score being higher than 80.

The median score of 85 indicates that half of the students scored 85 or above, and half 85 or below. This shows that the majority of the students performed well, with the middle score being in the high 80s.

The mode of 85 further reinforces that a significant number of student (3 out of 10) scored 85 on the quiz, which was the most common score.

Median. It is less affected by outliers or extreme values compared to the mean.

c) i. Mean, $\bar{x} = \frac{55+65+65+70+85+95+95+95+100+100}{10}$
 $= 82.5$

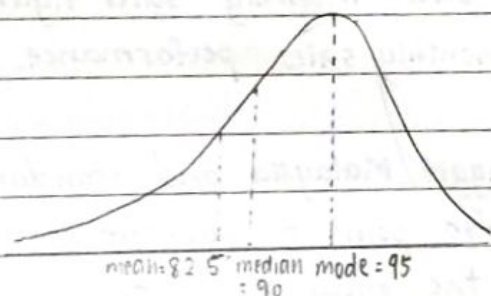
5th observation = 85

6th observation = 95

Median = $\frac{1}{2}(85+95)$
 $= 90$

Mode = 95

ii.



iii. In terms of median score, the exact performance of the students of this data is better compared to previous data.

Question 3

a) i. Range

$$\text{range} = 40 - 25 = 15$$

$$\bar{x} = \frac{25+30+28+35+32+27+40+38+33+36+31+29}{12} = 32$$

iii) standard deviation

$$s = \sqrt{19.167} = 4.378$$

ii) variance

$$s^2 = \frac{\sum_{i=1}^N (x_i - \bar{x})^2}{N}$$

$$= \frac{(25-32)^2 + (30-32)^2 + (28-32)^2 + (35-32)^2 + (32-32)^2 + (27-32)^2 + (40-32)^2 + (38-32)^2 + (33-32)^2 + (36-32)^2 + (31-32)^2 + (29-32)^2}{12}$$

$$= \frac{230}{12}$$

$$= 19.167$$

b) - The range of sales is 15, which the difference between the highest and lowest sales over the past year is RM15000, indicates a extreme different for the highest and lowest monthly sales.

The variance is a measurement of how far each number in a data set is from the mean.

Variance of 20.909 is a high variance, indicates that monthly sales has higher variability and the monthly sales performance is generally further from the mean.

The standard deviation helps to determine the spread of sales from their average sale, which is mean. Standard deviation of 4.573 is considered high, which means sales are more spread out.

Thus, the business's monthly sales performance is relatively inconsistent.

c) Understanding the variability in business's monthly sales performance throughout the year can target at those lower sales month, such as January, February, March, June and December, which are all lower than the average and find out what causes then solve it with related strategic. For example, organise a promotions during months that show lower sales. By understanding those, company can take action to do financial planning or marketing management to overcome the sales of monthly that relatively low.

4. $X \sim N(50, 10)$

(a) $X=50 \quad P(X > 50) = P\left(Z > \frac{50-50}{10}\right)$

$$= P(Z > 0)$$

$$= 0.5 - 0$$

$$= 0.5 \times 100\%$$

$$= 50\%$$

(L)

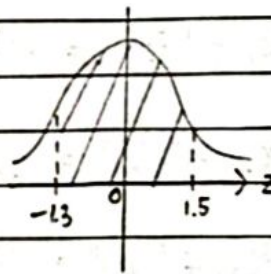
(b) $P(37 < X < 65) = P\left(\frac{37-50}{10} < Z < \frac{65-50}{10}\right)$

$$= P(-1.3 < Z < 1.5)$$

$$= P(Z < 1.5) + P(Z < 1.3)$$

$$= 0.4332 + 0.4082$$

$$= 0.8364$$



(c) $P(X < 20) = P\left(Z < \frac{20-50}{10}\right)$

$$= P(Z < -3)$$

$$= 0.5 - P(Z < 3)$$

$$= 0.5 - 0.4987$$

$$= 0.0013$$

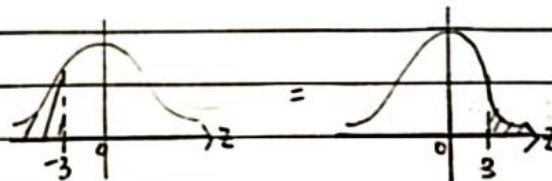
$$\text{Budget} = 2 \times \text{RM}200$$

$$= \text{RM}400$$

$$\text{Total person} = 0.0013 \times 1000$$

$$= 1.3 \approx 2$$

$$= 2$$



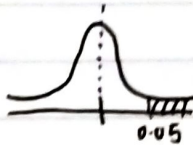
$$(d) \quad 5\% = 0.05$$

$$0.5 - 0.05 = 0.45$$

$$P(Z > \frac{m-50}{10}) = 0.45$$

$$\frac{m-50}{10} = 1.645$$

$$m = 66.45$$



5. (a) X is discrete random variable which represents the number of correct answers a student gets on the exam.

$$X = 0, 1, 2, 3, 4, 5, 6.$$

(b)	$X=x$	0	1	2	3	4	5	6
	$P(X=x)$	$\frac{729}{4096}$	$\frac{729}{2048}$	$\frac{1215}{4096}$	$\frac{135}{1024}$	$\frac{135}{4096}$	$\frac{9}{2048}$	$\frac{1}{4096}$

$$P(X=0) = {}^6C_0 \left(\frac{1}{4}\right)^0 \left(\frac{3}{4}\right)^6 = \frac{729}{4096}$$

$$P(X=1) = {}^6C_1 \left(\frac{1}{4}\right)^1 \left(\frac{3}{4}\right)^5 = \frac{729}{2048}$$

$$P(X=2) = {}^6C_2 \left(\frac{1}{4}\right)^2 \left(\frac{3}{4}\right)^4 = \frac{1215}{4096}$$

$$P(X=3) = {}^6C_3 \left(\frac{1}{4}\right)^3 \left(\frac{3}{4}\right)^3 = \frac{135}{1024}$$

$$P(X=4) = {}^6C_4 \left(\frac{1}{4}\right)^4 \left(\frac{3}{4}\right)^2 = \frac{135}{4096}$$

$$P(X=5) = {}^6C_5 \left(\frac{1}{4}\right)^5 \left(\frac{3}{4}\right)^1 = \frac{9}{2048}$$

$$P(X=6) = {}^6C_6 \left(\frac{1}{4}\right)^6 \left(\frac{3}{4}\right)^0 = \frac{1}{4096}$$

(c) $p = \frac{1}{4}, q = \frac{3}{4}, n = 6$

(d) $P(X \geq 3) = P(X=3) + P(X=4) + P(X=5) + P(X=6)$

$$\begin{aligned} \mu &= np \\ &= 6 \left(\frac{1}{4}\right) \\ &= 1.5 \end{aligned}$$

$$\begin{aligned} &= \frac{135}{1024} + \frac{135}{4096} + \frac{9}{2048} + \frac{1}{4096} \\ &= 0.1694 \end{aligned}$$

(e) The probability p of getting a question wrong is $1 - \frac{1}{4} = \frac{3}{4}$.

$$\begin{aligned} P(Y=4) &= (1-0.75)^{4-1} \times 0.75 \\ &= (0.25)^3 (0.75) \\ &= 0.012 \end{aligned}$$

6-(a) The probability distribution of X is negative binomial distribution which represents the number of customers that will arrive until the 4th customer who orders a cappuccino

(b) Standard deviation, $\sigma = \sqrt{\frac{r \cdot (1-p)}{p^2}}$

$$= \sqrt{\frac{4(1-0.70)}{0.7^2}}$$

$$\sigma = 1.565$$

(c) $P(X=6) = \binom{6-1}{4-1} (0.70)^4 (0.30)^{6-4}$

$$= \binom{5}{3} (0.70)^4 (0.30)^2$$
$$= 0.216$$

(d) $P(X=7) = \binom{7-1}{4-1} (0.70)^4 (0.30)^{7-4}$

$$= 0.158$$